A General Framework for Incorporating Identification Uncertainty in Individualized Treatment Rules

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Running Head: Incorporating Uncertainty in Treatment Rules

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Abstract

Estimating individualized treatment rules (ITRs) from observational data or clinical trials with non-adherence is challenging due to potential unmeasured confounding bias. Instrumental variables (IVs) can offer partial identification of the possible values for conditional average treatment effects (CATEs). However, optimal treatment decisions under partially identified CATEs can be uncertain; and current literature fails to inform such uncertainty in treatment decisions. In this work, we adopt the concept of a deferred treatment option and develop a novel class of 'optimal' ITRs called Enhanced IV-optimal ITRs with deferred option (EIV-ITRs) to guide treatment decisions and account for identification uncertainty. The deferred option extends beyond the original treatment options, identifying patients susceptible to identification uncertainty and allowing for the collection of additional information or physicians' guidance to aid in optimal decisions. Our framework allows users to control the size of subgroups receiving the deferred option by considering the relative cost of gathering additional information to ascertain decisions or the potential risk of taking the deferred option versus the cost of an incorrect treatment decision. To estimate the EIV-ITR, we develop a weighted classification framework with a modified hinge loss function, where the weights are non-smooth transformations of nuisance parameters. We further propose an augmented empirical risk minimization approach to estimate the EIV-ITRs, achieving a fast convergence rate even if the nuisance parameters are estimated using nonparametric or machine learning methods.

Keywords: Individualized treatment rules, Identification uncertainty, Instrumental variable, Partial identification, Deferred option

1 Introduction

1.1 Estimating individualized treatment rules under unmeasured confounding bias

Comparing candidate treatment options based on their effects on an outcome of interest is common in clinical practice. When treatment effects vary among individuals, individualized treatment strategies that assign treatments based on individual characteristics can be more beneficial. Individualized treatment strategies use an individualized treatment rule (ITR) to map characteristics to specific treatment options, typically optimizing the mean outcome, resulting in an optimal ITR. The estimation of optimal ITRs has been extensively studied, with contributions from various researchers (Murphy 2003; Robins 2004; Zhao et al. 2012; Zhang et al. 2012a,b; Chakraborty et al. 2014; Laber and Zhao 2015; Luedtke and van der Laan 2016; Tao et al. 2018; Athey and Wager 2021). Q-learning and A-learning have been proposed to estimate the contrast function or conditional average treatment effect (CATE) for constructing the ITR (Watkins and Dayan 1992; Chakraborty et al. 2010; Qian and Murphy 2011; Laber et al. 2014; Lu et al. 2013; Shi et al. 2016, 2018). A classification-based framework by Zhao et al. (2012) allows direct estimation of optimal ITRs and has been extended to robust estimation and inference (Zhao et al. 2019; Liang et al. 2022). For a comprehensive review of various methods, refer to Moodie et al. (2007); Chakraborty and Murphy (2014); Schulte et al. (2014).

Estimating ITRs can be challenging when unmeasured confounding bias exists. Unmeasured covariates may correlate both the treatment received by patients and their outcomes. The issue of treatment recommendation in the presence of unmeasured confounding has recently gained attention (Qiu et al. 2021; Qi et al. 2023; Cui and Tchetgen 2021; Cui and Tchetgen Tchetgen 2021). When an instrumental variable (IV) is observable in the data, one solution to address possible unmeasured confounding is through partial identification of CATE (Angrist et al. 1996; Balke and Pearl 1997;

Manski 2003; Robins and Greenland 1996). For a comprehensive review of the partial identification of average treatment effect (ATE) or CATE using an IV, we refer readers to Swanson et al. (2018). In learning ITRs under partially identified CATEs, Pu and Zhang (2021a) proposed minimizing the risk in the worst scenario. However, the resulting ITR would still assign a treatment option to patients whose optimal decisions are uncertain due to the partial identification of CATE, without alerting these patients to the possible identification uncertainty in their treatment decisions.

In safety-critical applications, such as treatment recommendations, incorrect decisions can result in severe consequences for patients. Informing when the ITR may be susceptible to identification uncertainty can be as important as improving its average or overall performance. In classification literature, a reject option is adopted in estimated decisions rules to abstain from making predictions in uncertain cases (Chow 1957, 1970; Fumera and Roli 2002; Fumera et al. 2003, 2000; Hansen et al. 1997; Herbei and Wegkamp 2006; Bartlett and Wegkamp 2008; Wegkamp and Yuan 2011; Yuan and Wegkamp 2010). Through incorporating the reject option, the reject option informs the uncertain cases, and for those not taking the reject option, the decision seems to be less susceptible to the underlying uncertainty. For ITR learning, we recast this concept and refer to it as the deferred option. Different from the reject option, a deferred decision indicates uncertain treatment decisions from the available data and urges treatment decisions guided by other clinical considerations and patient preferences. Similar considerations exist in Yuan (2015) assuming point identification of CATEs. The resulting ITR takes deferred option for those whose relative contrast functions do not show large benefits. However, in the presence of identification uncertainty, small benefits do not necessarily indicate uncertain optimal decisions. Therefore, whether and how we can effectively link a deferred option to inform identification uncertainty remains an open question.

Our contribution lies in establishing a link between a deferred option and identification uncertainty in estimating ITRs. Our novel framework simultaneously estimates optimal treatment decisions for the subgroup with identifiable optimal treatment decisions, and informs those susceptible to identifi-

cation uncertainty using a deferred option. For patients receiving the deferred option, their treatment decisions may follow other clinical considerations (see Remark 2). Users can balance the trade-off between the cost (or the risk) of possible alternative strategies and the risk of making incorrect treatment decisions only based on the data available. Further, to estimate the optimal ITR within this framework, we adopt an augmented empirical risk minimization (ERM) approach. In the estimation, we adopt a weighted classification framework with a modified hinge loss function (Wegkamp and Yuan 2011). The weights are obtained through a piecewise-linear transformation of the upper and lower bounds of the partially identified CATEs. Nonparametric or machine learning methods are used to estimate these bounds and construct the weights. To reduce the finite-sample bias from these methods, we construct augmentation terms that approximate the first order bias of these methods and then remove the estimated bias from the proposed loss function. By minimizing the augmented empirical risk, we estimate the optimal ITR with a deferred option. In our theoretical investigation, we introduce generalized geometric conditions, and investigate the approximation error when using Gaussian kernels with the deferred option; we further establish a fast convergence rate using augmented ERM compared to ERM without augmentations.

1.2 A motivating example

Our motivating example is the Prostate Cancer Prevention Trial (PCPT), a landmark clinical study conducted from 1993 to 2003 (Thompson et al. 2003). The PCPT was a double-blind, randomized trial carried out across multiple centers to investigate the impact of finasteride on prostate cancer prevention among over 18,000 initially healthy men. Thompson et al. (2003) reported a decreased risk of developing overall prostate cancer but an elevated risk of developing high-grade prostate cancer among participants. One follow-up research question is who will be safe or unsafe taking finasteride for cancer prevention.

These intriguing results have motivated researchers to explore the potential underlying expla-

nations. Pinsky et al. (2021) found that 16.5% of patients did not adhere to the study protocol, indicating that a patient receiving finasteride might not take the medication. Therefore, one direction is to explore whether there are differential effects associated with actually taking finasteride. Further, they showed that most non-adherence occurred in the finasteride arm, suggesting possible unmeasured confounding between actual treatments taken by patients and their outcomes. To infer the treatment effect based on whether patients take finasteride rather than their treatment assignments, we need a framework to assess the treatment effect in the presence of possible unmeasured confounding. One possible strategy is to use the IV framework. An IV is a variable that is correlated with the treatment but is independent of the unmeasured confounders that affect the outcome. This framework helps isolate the variation in treatment that is free from confounding and thus allows to infer treatment effects even when there is possible unmeasured confounding. In the PCPT, the initial treatment assignment, being randomized, serves as a natural IV. Using this IV, our proposed method can inform who will be safe or unsafe if, indeed, taking finasteride for cancer prevention, as well as who faces uncertain risks due to unmeasured confounding bias, and thus other clinical considerations and patient preferences are needed for treatment decisions.

Remark 1. *In this work, our method targets the causal effect of treatment receipt (finasteride use), rather than the intention-to-treat effect of assignment. IVs are used to address unmeasured confounding and partially identify this effect. The resulting ITR therefore recommends the treatment that would be optimal if taken; whether a patient ultimately adheres is a separate behavioral question. Consequently, non-compliance does not undermine the interpretation of the learned ITR. Moreover, the deferred option guards against overconfident recommendations when identification remains weak due to non-compliance. Unlike intention-to-treat analyses, which base recommendations on assigned treatment regardless of adherence, our approach evaluates heterogeneity from an efficacy perspective. This is an important viewpoint for patient-centered decision-making, where interest lies in the true effect of the treatment itself rather than an average effect diluted by others' non-adherence.*

2 IV-optimal ITR with a deferred option

We propose our framework incorporating a deferred option in ITR to account for identification uncertainty when using an IV. First, we discuss the partial identification of CATEs under IV assumptions, highlighting the challenges using partial identification to inform optimal treatment decisions. Next, we present a general framework for IV-optimal ITRs with a deferred option, outlining how this approach can improve decision-making by identifying cases with uncertain optimal treatment decisions, especially those whose risk of making wrong treatment decisions is relatively higher than taking the deferred option.

2.1 Partial identification of CATEs

Let $\boldsymbol{X} \in \mathcal{X}$ be a p-dimensional covariate vector, and let $A \in \{-1, 1\}$ be a binary variable indicating the treatment option taken by the patient. Denote the observed outcome of interest as Y. Under the potential outcome framework, we define Y_a as the potential outcome given $A = a$. Given any ITR, $D(\boldsymbol{X}) : \mathcal{X} \to \{-1, 1\}$, when CATEs are identifiable for each individual, the value function $V(D) = E\left[Y_{D(\boldsymbol{X})}\right]$ is also identifiable. Thus, the optimal ITR, which maximizes the value function, can be identified. The optimal ITR follows the sign of the CATE to assign treatment to each patient, i.e., $D^*(\boldsymbol{X}) = \operatorname{sgn}\{\Delta(\boldsymbol{X})\}$, where $\Delta(\boldsymbol{X}) = E\left[Y_1 - Y_{-1} \mid \boldsymbol{X}\right]$ is the CATE, and $\operatorname{sgn}(t) = 1$ if $t \geqslant 0$; and $\operatorname{sgn}(t) = -1$, otherwise.

However, the identifiability of CATEs requires assumptions, such as the absence of unmeasured confounding, which can be violated in many applications (see our motivating example). An IV is a useful tool to infer CATEs, especially when there is possible unmeasured confounding between the treatment (e.g., actual treatment taken by the patients) and outcomes. Let $Z \in \{-1, 1\}$ be a binary IV. Under the potential outcome framework, we define A_z as the indicator of whether the patient would take the treatment or not under the assignment $Z = z$, and $Y_{z,a}$ as the potential outcome given

the treatment $Z = z$ and assignment $A = a$. Under this framework, we assume the following core IV conditions (Angrist et al. 1996):

1. Exclusion restriction: $Y_{-1,a} = Y_{1,a}$ for any a, and thus we can use Y_a to denote them;
2. Consistency: $Z = z$ implies that $A = A_z$; further, $A = a$ implies that $Y = Y_a$;
3. IV unconfoundedness: $Y_a, A_z \perp\!\!\!\perp Z \mid \boldsymbol{X}$;
4. Positive correlation between IV and treatment: $P(A = 1 \mid Z = 1, \boldsymbol{X}) > P(A = 1 \mid Z = -1, \boldsymbol{X})$ almost surely.

Under these IV assumptions, the CATE (or ATE) is not point identifiable. However, they result in identifiable bounds on the CATE. Under these assumptions, we can identify non-trivial functions $L(\boldsymbol{X})$ and $U(\boldsymbol{X})$ such that $L(\boldsymbol{X}) \leqslant \Delta(\boldsymbol{X}) \leqslant U(\boldsymbol{X})$. For binary outcomes, these IV assumptions lead to the well-known Balke-Pearl bound (Balke and Pearl 1997). For a continuous bounded outcome, Manski and Pepper (2000) proposed to use a monotone IV assumption and derived a nonparametric bound for the ATE (or CATE). Further discussions can be found in Manski (2003); Manski and Pepper (2009).

Under additional assumptions, we may be able to shorten these bounds or to identify the ATE in an identifiable group or on the entire population. For example, Richardson and Robins (2010) proposed to bound the ATE as a function of the defier proportion. As a special case, when there are no defiers, a valid IV can be used to identify the ATE among the compliers (Angrist et al. 1996). More recently, Swanson et al. (2018) discussed various proposals on how to bound ATE through setting bounds to $E[Y_{-1}]$ among always-takers and $E[Y_1]$ among never-takers. Wang and Tchetgen Tchetgen (2018); Cui and Tchetgen Tchetgen (2021); Qiu et al. (2021) proposed assumptions that lead to point identification of ATE or CATE. In these investigations, the proposed assumptions, e.g., Assumption 5(a) or 5(b) in Wang and Tchetgen Tchetgen (2018); Assumption 7 in Cui and Tchetgen Tchetgen (2021); and Assumption 5(b) in Qiu et al. (2021), may require that the unmeasured confounding factors cannot

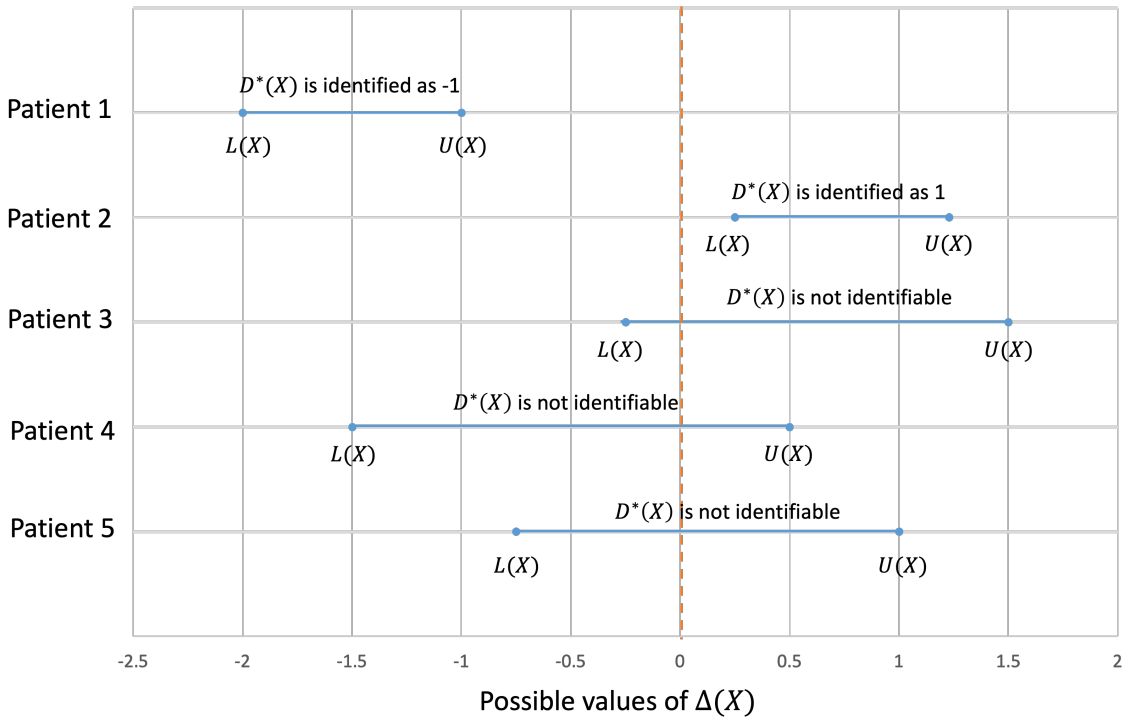


Figure 1: Optimal treatment decision is identifiable for Patients 1 and 2 but not identifiable for Patients 3, 4, and 5.

modify the CATE, which contradicts the underlying belief of precision medicine, where many measured or unmeasured factors may contribute to treatment effect heterogeneity. These are not the focus of this work; our proposed framework can extend to other bounds or point identified CATEs.

Based on the Balke-Pearl bound (Balke and Pearl 1997), denoted as $L(\boldsymbol{X})$ and $U(\boldsymbol{X})$, the optimal treatment decisions might be identifiable. Figure 1 illustrates some examples. If the bound does not contain 0, i.e., $L(\boldsymbol{x})U(\boldsymbol{x}) \geqslant 0$, then the optimal treatment decision for this patient is identifiable. However, when the bound contains 0, i.e., $L(\boldsymbol{x})U(\boldsymbol{x}) < 0$, then the optimal treatment decision is not identifiable since $\Delta(\boldsymbol{X})$ can be negative in some cases and positive in others.

2.2 A general framework for IV-optimal ITR with a deferred option

When $\Delta(\boldsymbol{X})$ is point identifiable, the optimal ITR D^* can be characterized by the concept of total loss in value. The total loss in value function is the average of the loss in value if the ITR makes an incorrect decision w.r.t. D^* over the entire population. Given an ITR, D, which maps covariates

to -1 or 1, for each individual, if the treatment received follows the optimal ITR $D^*(\boldsymbol{X})$, the loss in value for this individual is 0; if the treatment receipt is not the optimal treatment decision, the loss in value for this individual is $|\Delta(\boldsymbol{X})|$, which is the difference in the outcome under the optimal ITR vs. the given ITR. Thus, the total loss in value is $E\left[TLV(D, \boldsymbol{X}; \Delta)\right]$, where $TLV(D, \boldsymbol{X}; \Delta) \equiv |\Delta(\boldsymbol{X})| \, 1\left\{D(\boldsymbol{X}) \neq \operatorname{sgn}\left\{\Delta(\boldsymbol{X})\right\}\right\}$ is the individual-level loss in value. When $\Delta(\boldsymbol{X})$ is point identifiable, minimizing the total loss in value leads to the optimal ITR D^*.

When $\Delta(\boldsymbol{X})$ is only partially identified, we augment the individual-level loss in value with the loss of a deferred option. The deferred option is an option to abstain from making any treatment decision only based on the data available in decision-making. In this work, the deferred option is denoted by 0. We define an ITR with a deferred option as a mapping from the space of covariates $\mathcal{X}$ to the extended treatment space including the deferred option $\{-1, 1, 0\}$. Given an ITR with a deferred option D and a patient $\boldsymbol{X}$, if $D(\boldsymbol{X}) = 0$, we defer making any treatment decision for this patient only based on the data available, and instead, we suggest other clinical considerations and patient preferences (see Remark 2). Under an ITR with a deferred option, for each individual, we then consider loss in value with a deferred option as

$$\widetilde{TLV}(D, \boldsymbol{X}; \Delta, d) = TLV(D, \boldsymbol{X}; \Delta) 1\{D(\boldsymbol{X}) \neq 0\} + d|\Delta(\boldsymbol{X})| 1\{D(\boldsymbol{X}) = 0\},$$

where $d \in (0, 1]$ is a discount factor. We further define the total loss in value with a deferred option as $E\left[\widetilde{TLV}(D, \boldsymbol{X}; \Delta, d)\right]$. Compared with $TLV(D, \boldsymbol{X}; \Delta)$, the loss in value with a deferred option has an additional term $d|\Delta(\boldsymbol{X})|1\{D(\boldsymbol{X}) = 0\}$, which represents the risk of following the deferred option. Notice that when $\Delta(\boldsymbol{X})$ is point identified, $|\Delta(\boldsymbol{X})|$ is the loss if the treatment received is not the optimal treatment decision. The discount factor d accounts for the relative loss (or risk) of following the deferred option vs. not making the optimal treatment decision. Other formulations for $d|\Delta(\boldsymbol{X})|$ may be possible and yield the same resulting ITR with a deferred option; we choose this formulation

for its simplicity. In practice, $d|\Delta(\boldsymbol{X})|$ is interpreted as the loss in value if choosing the deferred option. A wrong treatment assignment has a loss of $|\Delta(\boldsymbol{X})|$; deferral, e.g., extra testing, specialist review, or waiting for additional information, is assumed to incur only a fraction d of the cost. When user specifies a choice d based on the cost of deferral, $d|\Delta(\boldsymbol{X})|$ is an estimate of the loss in clinical utility with the deferred option. When d is specified following other criteria (see Remark 4), d is just a tuning parameter to meet the specified need, e.g., certain proportion of deferrals/high deferral performance, rather than an estimate of actual clinical utility.

Remark 2. *A deferred decision does not mean that treatment should be withheld. Rather, it indicates that, for the primary outcome under consideration, the causal effect of actually taking the treatment is not reliably identified from the available data. In this case, additional clinical considerations may provide complementary information to support individualized decision-making. For example, for patients assigned to the deferred option, treatment can still be given. We recommend to consider the following alternatives: 1) default to standard care, 2) use patient preferences, and/or 3) other clinically relevant considerations such as toxicity tradeoffs. When the primary outcome can lead to identifiable optimal treatment decisions, our treatment decisions are still prioritized with respect to the primary outcome deemed most clinically relevant. If one ignores the deferred option and simply assigns treatment based on the point estimate of CATE ignoring the potential confounding bias, the resulting decision may be driven by the confounding rather than true treatment effect heterogeneity.*

The proposed total loss in value with a deferred option is an extension of the (total) loss in value when $\Delta(\boldsymbol{X})$ is identifiable. We can show that for any choice of $d \in (0, 1]$, when $\Delta(\boldsymbol{X})$ is identifiable, minimizing the proposed (total) loss in value with a deferred option over all possible ITRs leads to the optimal ITR $D^*(\boldsymbol{X})$, which never takes the deferred option. Thus, when $\Delta(\boldsymbol{X})$ is identifiable, minimizing the total loss in value without any deferred option is equivalent to minimizing the proposed total loss in value with a deferred option.

It is likely to make an incorrect treatment decision when $\Delta(\boldsymbol{X})$ is not fully identifiable. This occurs

when the interval $[L(\boldsymbol{X}), U(\boldsymbol{X})]$ contains 0, indicating that $\Delta(\boldsymbol{X})$ could take on a range of possible values including both positive and negative values under identification uncertainty. In this case, our proposed (total) loss in value with a deferred option enables a trade-off between the (worst) inevitable risk of making incorrect treatment decisions vs. the risk of following a deferred option. By choosing an appropriate level of d, we control the acceptable risk of opting for a deferred decision versus making incorrect treatment decisions. A lower value of d indicates a low risk of following a deferred option. In other words, we have a high tolerance for following a deferred option, where we tend to defer decisions to other clinical considerations and patients' preferences in making a deterministic treatment decision whenever there is any risk of making incorrect decisions due to identification uncertainty. Conversely, a higher value of d implies a high risk for following a deferred option, allowing us to make treatment decisions between 1 and -1 as the risk of making incorrect treatment decisions is low. When $d = 1$, we can demonstrate that the deferred option is never taken unless $L + U = 0$, which corresponds to a symmetric bound for $\Delta(\boldsymbol{X})$ around 0. In this scenario, the proposed total loss in value with a deferred option aligns with the loss proposed in Pu and Zhang (2021a).

Under the proposed (total) loss in value with a deferred option, we define the IV-optimal ITR with a deferred option, as

$$D_d^*(\boldsymbol{X}) = \arg\min_{D} \sup_{\Delta(\boldsymbol{X})\in[L(\boldsymbol{X}),U(\boldsymbol{X})]} E\left[\widetilde{TLV}(D, \boldsymbol{X}; \Delta, d)\right]. \tag{1}$$

The following theorem provides characterizations of the IV-optimal ITR with a deferred option.

Theorem 1. *Define*

$$D_d^*(\boldsymbol{X}) = \begin{cases} 1 & L_-(\boldsymbol{X}) < dU_+(\boldsymbol{X}), \\ -1 & U_+(\boldsymbol{X}) < dL_-(\boldsymbol{X}), \\ 0 & \textit{otherwise,} \end{cases}$$

where $L_-(\boldsymbol{X}) = |L(\boldsymbol{X})|\,1\{L(\boldsymbol{X}) \leqslant 0\}$ *and* $U_+(\boldsymbol{X}) = |U(\boldsymbol{X})|\,1\{U(\boldsymbol{X}) \geqslant 0\}$. *The* D_d^* *is an IV-optimal ITR with a deferred option, which minimizes the worst total loss in value with a deferred option under the discount factor* d.

Notably, when $d \to 0$, our proposed IV-optimal ITR with a deferred option tends to always chose the deferred option for patients whose optimal treatment decision is unidentifiable due to identification uncertainty of $\Delta(\boldsymbol{X})$. On the other hand, when $d = 1$, the IV-optimal ITR coincides with the IV-optimal ITR proposed in Pu and Zhang (2020). For all possible choices of d, the proposed IV-optimal ITR with a deferred option follows the optimal ITR $D^*(\boldsymbol{X})$ for patients whose optimal treatment decision is indeed identifiable. For simplicity, we call our framework as EIV-ITR, which Enhances the IV-optimal ITR with a deferred option.

Remark 3. *To further explain* $D_d^*(\boldsymbol{X})$, *we focus on the case when* $L(\boldsymbol{X})U(\boldsymbol{X}) < 0$ *(e.g., Patients 3, 4, 5 in Figure 1). When* $L(\boldsymbol{X})U(\boldsymbol{X}) < 0$, *the value of* $D_d^*(\boldsymbol{X})$ *depends on the ratio* $L_-(\boldsymbol{X})/U_+(\boldsymbol{X})$, *which is the ratio of the distances from* $L(\boldsymbol{X})$ *and* $U(\boldsymbol{X})$ *to* 0. *When* $L(\boldsymbol{X})$ *is closer to* 0 *compared with* $U(\boldsymbol{X})$, *i.e.,* $|L(\boldsymbol{X})|/|U(\boldsymbol{X})| < d$, *the* $D_d^*(\boldsymbol{X})$ *suggests that the patient should take the treatment; When* $U(\boldsymbol{X})$ *is closer to* 0 *compared with* $L(\boldsymbol{X})$, *i.e.,* $|U(\boldsymbol{X})|/|L(\boldsymbol{X})| \leqslant d$, *the* $D_d^*(\boldsymbol{X})$ *suggests that the patient should take the control; otherwise, the* $D_d^*(\boldsymbol{X})$ *suggests that the patient should abstain from making any treatment decisions, and clinicians should collect additional information to aid in making optimal decisions. Figure 2 illustrates the* $D_d^*(\boldsymbol{X})$ *with* $d = 1/3$.

Remark 4. *The choice of* d *determines the proportion of patients following the deferred option. When* d *is small, more patients will receive the deferred option. However, only patients whose treatment effects are susceptible to unmeasured confounding are subjected to the deferred option. The choice of* d *reflects the relative loss due to uncertainty in identifying optimal treatment decisions versus making a wrong treatment decision. Therefore, it should be controlled by users based on domain knowledge. When there is limited domain knowledge, we can implement the proposed method with multiple* d*'s or calculate the skewness of the partial identification interval for each patient, i.e.,*

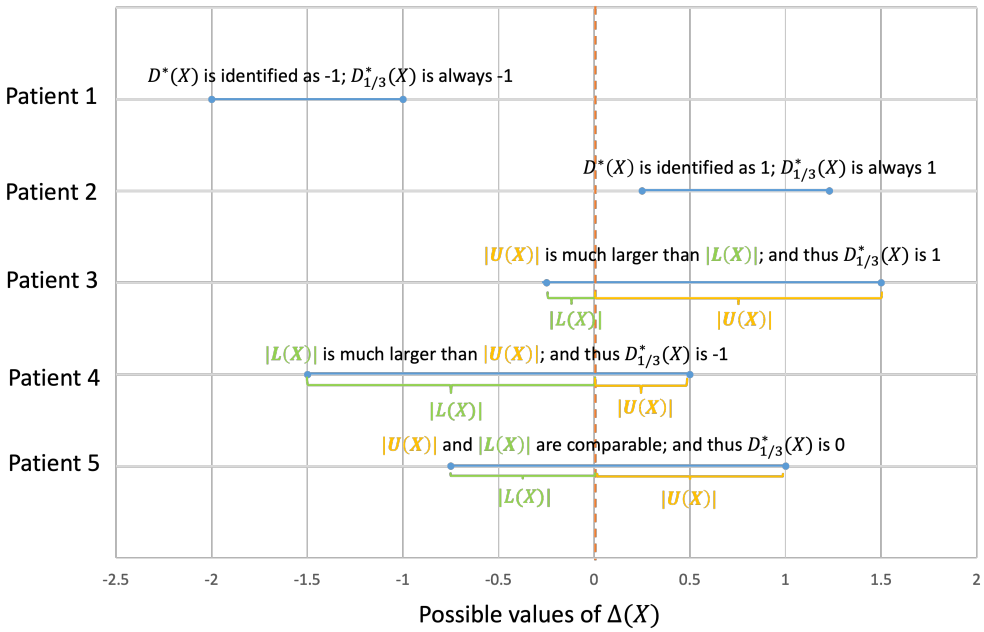


Figure 2: Illustration of $D^*_d(\boldsymbol{X})$ with $d = 1/3$ for five individuals. The $D^*_{1/3}(\boldsymbol{X})$ suggests the deferred option if and only if $L(\boldsymbol{X})U(\boldsymbol{X}) < 0$ and $|L(\boldsymbol{X})/U(\boldsymbol{X})| \in (1/3, 3)$, i.e., $|L(\boldsymbol{X})|$ and $|U(\boldsymbol{X})|$ are comparable.

$\min\{\widehat{U}_+(\boldsymbol{X})/\widehat{L}_-(\boldsymbol{X}), \widehat{L}_-(\boldsymbol{X})/\widehat{U}_+(\boldsymbol{X})\}$. *Given a specific choice of d, our proposed method assigns the deferred option to patients with skewness greater than d. Thus, given a prespecified desired proportion of deferred options, we can select d to achieve such proportion of deferred options. Another choice is to choose d based on the proportion of deferred subjects with treatment-effect bounds crossing zero, which is a metric to evaluate the performance of deferral. Higher proportion of deferred subjects with treatment-effect bounds crossing zero indicates that we defer correctly for patients with unidentified optimal treatments. In implementation, the proportion of deferred subjects with treatment-effect bounds crossing zero can be estimated using the cross-validation.*

3 Augmented empirical risk minimization

In this section, we first propose a weighted classification framework, which includes the use of a modified hinge loss function to handle deferred option. Then, we introduce an augmented empirical risk minimization (ERM) approach to improve estimation of the EIV-ITR.

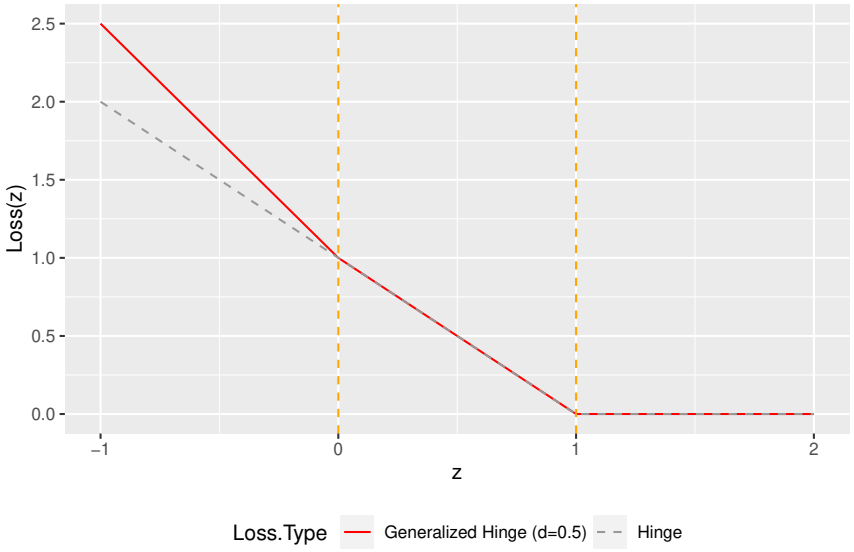


Figure 3: Comparison between $\phi_{0.5}(z)$ and $\phi_1(z)$ (hinge loss).

3.1 Estimation via a convex surrogate loss

To estimate the EIV-ITR, we construct a convex surrogate loss function. Following Bartlett and Wegkamp (2008), we utilize a generalized hinge loss function as a surrogate loss (see Figure 3), denoted as $\phi_d(z)$, which is defined as follows:

$$\phi_d(z) = \begin{cases} 1 - z/d & \text{if } z < 0, \\ 1 - z & \text{if } 0 \leqslant z < 1, \\ 0 & \text{otherwise.} \end{cases}$$

Different from the standard hinge loss, the generalized hinge loss introduces an additional non-differentiable point at $z = 0$. The intuition is that the deferred option can be considered as a third treatment option, and the additional non-differentiable point at $z = 0$ accommodates this additional treatment option.

Motivated by this, we construct the following surrogate loss function

$$\min_f E\left[U_+(X)\phi_d(f) + L_-(X)\phi_d(-f)\right].$$

Let the minimizer of this surrogate loss be denoted as f_d^*. Then we transform the f_d^* into the EIV-ITR by

$$D\left(\boldsymbol{X}; f, \delta\right) = \begin{cases} 1 & f(\boldsymbol{X}) > \delta, \\ -1 & f(\boldsymbol{X}) \leqslant -\delta, \\ 0 & \text{otherwise}, \end{cases}$$

where δ is a non-negative constant. In Section 4, we show that we can estimate the EIV-ITR by first estimating f_d^*, which minimizes the proposed surrogate loss, and then constructing the ITR using $D(\boldsymbol{X}; f, \delta)$.

To estimate f_d^*, we first estimate $L(\boldsymbol{X})$ and $U(\boldsymbol{X})$ and then apply traditional ERM. Using the Balke-Pearl bound introduced in Balke and Pearl (1997), we have

$$L(X) = \max\left\{P_1(\boldsymbol{X}), \cdots, P_8(\boldsymbol{X})\right\}, \quad U(X) = \min\left\{Q_1(\boldsymbol{X}), \cdots, Q_8(\boldsymbol{X})\right\},$$

where $P_k(\boldsymbol{X})$'s and $Q_k(\boldsymbol{X})$'s are linear transformations of $p_{y,a|z}(\boldsymbol{X})$ with $p_{y,a|z}(\boldsymbol{X}) \equiv P(Y = y, A = a \mid Z = z, \boldsymbol{X})$. Explicit formulas for the Balke-Pearl bound can be found in the online Supplementary Materials. To estimate $L(\boldsymbol{X})$ and $U(\boldsymbol{X})$, we can use nonparametric or machine learning methods to estimate $p_{y,a|z}(\boldsymbol{X})$'s. Let $\widehat{L}(\boldsymbol{X})$ and $\widehat{U}(\boldsymbol{X})$ be the estimates obtained by replacing $p_{y,a|z}(\boldsymbol{X})$'s with their estimates $\widehat{p}_{y,a|z}(\boldsymbol{X})$'s. We then use traditional ERM to estimate f_d^*. Let $\mathcal{H}$ be a pre-specified function class, e.g., reproducing kernel Hilbert space (RKHS). Traditional ERM solves the following objective function

$$E_n\left[\widehat{U}_+(\boldsymbol{X})\phi_d(f(\boldsymbol{X})) + \widehat{L}_-(\boldsymbol{X})\phi_d(-f(\boldsymbol{X}))\right] + \lambda_n \left\|f\right\|_{\mathcal{H}}^2 \tag{2}$$

over $f \in \mathcal{H}$, where the $\|\cdot\|_{\mathcal{H}}^2$ is the associated norm with $\mathcal{H}$ and λ_n is a tuning parameter. However, due to the adoption of nonparametric or machine learning methods to estimate $p_{y,a|z}(\boldsymbol{X})$'s, the convergence

rates of $\widehat{L}(\boldsymbol{X})$ and $\widehat{U}(\boldsymbol{X})$ are slow, which can result in poor estimation of f_d^*. To overcome the slow convergence rates caused by these estimates, we adopt augmentations terms to remove the first-order bias of these estimates and achieve a faster convergence rate.

3.2 Improved estimation via augmentations

In this section, we propose an augmented ERM procedure to improve the convergence rate of traditional ERM. Without loss of generality, we assume that

$$L(\boldsymbol{X}) = \max\left\{P_1(\boldsymbol{X}), \cdots, P_K(\boldsymbol{X})\right\}, \quad U(\boldsymbol{X}) = \min\left\{Q_1(\boldsymbol{X}), \cdots, Q_K(\boldsymbol{X})\right\},$$

where $P_k(\boldsymbol{X})$'s and $Q_k(\boldsymbol{X})$'s are smooth functions depending on $p_{y,a|z}(\boldsymbol{X})$'s. As discussed in Section 3.1, using nonparametric or machine learning methods to estimate $p_{y,a|z}(\boldsymbol{X})$'s and directly solve the traditional ERM with plugg-in estimates, the estimation is susceptible to the slow convergence rates of the nonparametric or machine learning methods.

To overcome the slow convergence rates, we construct bias functions, e.g., $B_+(\boldsymbol{X}, Y, A, Z)$ and $B_-(\boldsymbol{X}, Y, A, Z)$, to estimate the bias of $\widehat{L}(\boldsymbol{X})$ and $\widehat{U}(\boldsymbol{X})$ w.r.t $L(\boldsymbol{X})$ and $U(\boldsymbol{X})$. These bias estimates are then used to augment the traditional ERM, resulting in the following augmented objective function:

$$E_n\left[\left\{\widehat{U}_+(\boldsymbol{X}) - B_+(\boldsymbol{X}, Y, A, Z)\right\}\phi_d(f(\boldsymbol{X})) + \left\{\widehat{L}_-(\boldsymbol{X}) - B_-(\boldsymbol{X}, Y, A, Z)\right\}\phi_d(-f(\boldsymbol{X}))\right] + \lambda_n \|f\|_{\mathcal{H}}^2. \quad (3)$$

To construct $B_+(\boldsymbol{X}, Y, A, Z)$ and $B_-(\boldsymbol{X}, Y, A, Z)$, we investigate the first-order bias of the traditional ERM objective function, considering the estimation errors of $\widehat{p}_{y,a|z}(\boldsymbol{X})$ with respect to $p_{y,a|z}(\boldsymbol{X})$. We consider the following approximation,

$$\widehat{U}_+(\boldsymbol{X}) - U_+(\boldsymbol{X}) \approx \sum_{y,a,z} \partial_{y,a,z} U_+(\boldsymbol{X}) \left\{\widehat{p}_{y,a|z}(\boldsymbol{X}) - p_{y,a|z}(\boldsymbol{X})\right\},$$

where $\partial_{y,a,z}U_+(\boldsymbol{X}) = -1\{U_+(\boldsymbol{X}) \geqslant 0\}\sum_{k=1}^{K} 1\left\{Q_k(\boldsymbol{X}) \leqslant \min_{k'\neq k} Q_{k'}(\boldsymbol{X})\right\}\partial_{y,a,z}Q_k(\boldsymbol{X})$, and $\partial_{y,a,z}Q_k(\boldsymbol{X})$ is the partial derivative of $Q_k(\boldsymbol{X})$ w.r.t $p_{y,a|z}(\boldsymbol{X})$. To estimate this approximation, we use the inverse probability weighting estimator, i.e.,

$$B_+(\boldsymbol{X}, Y, A, Z) = \sum_{y,a,z} \partial_{y,a,z}\widehat{U}_+(\boldsymbol{X})\frac{1\{Z=z\}}{\widehat{\pi}_z(\boldsymbol{X})}\left(\widehat{p}_{y,a|z}(\boldsymbol{X}) - 1\{Y=y, A=a\}\right),$$

where $\partial_{y,a,z}\widehat{U}_+(\boldsymbol{X}) = 1\left\{\widehat{U}_+(\boldsymbol{X}) \geqslant 0\right\}\sum_{k=1}^{K} 1\left\{\widehat{Q}_k(\boldsymbol{X}) \leqslant \min_{k'\neq k} \widehat{Q}_{k'}(\boldsymbol{X})\right\}\partial_{y,a,z}\widehat{Q}_k(\boldsymbol{X})$, and $\widehat{\pi}_z(\boldsymbol{X})$ is the estimator for $P(Z=z \mid \boldsymbol{X})$. Similarly, the bias of $\widehat{L}_-(\boldsymbol{X}) - L_-(\boldsymbol{X})$ is approximated by

$$B_-(\boldsymbol{X}, Y, A, Z) = \sum_{y,a,z} \partial_{y,a,z}\widehat{L}_-(\boldsymbol{X})\frac{1\{Z=z\}}{\widehat{\pi}_z(\boldsymbol{X})}\left(\widehat{p}_{y,a|z}(\boldsymbol{X}) - 1\{Y=y, A=a\}\right),$$

where $\partial_{y,a,z}\widehat{L}_-(\boldsymbol{X}) = -1\left\{\widehat{L}_-(\boldsymbol{X}) \leqslant 0\right\}\sum_{k=1}^{K} 1\left\{\widehat{P}_k(\boldsymbol{X}) \geqslant \max_{k'\neq k} \widehat{P}_{k'}(\boldsymbol{X})\right\}\partial_{y,a,z}\widehat{P}_k(\boldsymbol{X})$.

With these bias estimates, we further adopt the cross-fitting procedure proposed in Chernozhukov et al. (2018) to account for the nonparametric or machine learning estimates and the non-smooth transformation in bias estimation. Details on the implementation can be found in the online Supplementary Materials. We denote the final estimator from this improved procedure as $\widehat{f}_d$. In Section 4, we show that using the cross-fitting procedure, the approximations can effectively remove the first-order bias, resulting in a faster convergence rate for the augmented ERM compared to traditional ERM.

4 Theoretical properties

In this section, we provide the theoretical properties of the proposed estimator $\widehat{D} = D(\boldsymbol{X}; \widehat{f}_d, \delta)$. We focus on the generalization error $R\left(\widehat{D}\right) - R^*$, where

$$R(D) \;=\; E\left[\sup_{\Delta(\boldsymbol{X})\in[L(\boldsymbol{X}),U(\boldsymbol{X})]} \widetilde{TLV}(D, \boldsymbol{X}; \Delta, d)\right], \quad R^* = \min_D R(D).$$

To this end, we provide 1) excess risk bounds, which connect the surrogate loss with $R(D)$ and inform possible choices of δ; 2) an upper bound of approximation errors when using the RKHS with Gaussian radial basis function (RBF) kernels as $\mathcal{H}$; and 3) an upper bound of the generalization error, which converges to 0 with a faster convergence rate if using augmented ERM vs. traditional ERM. In the main text, we focus on the excess risk bounds and the generalization error; the discussion on the approximation errors can be found in the online Supplementary Materials.

The excess risk bound aims to bound $R(D) - R^*$ using the excess risk of the surrogate loss, i.e. $R_{\phi_d}(f) - \min_f R_{\phi_d}(f)$, where $R_{\phi_d}(f) = E\left[U_+(\boldsymbol{X})\phi_d(f(\boldsymbol{X})) + L_-(\boldsymbol{X})\phi_d(-f(\boldsymbol{X}))\right]$, and $R^*_{\phi_d} = \min_f R_{\phi_d}(f)$. We have the following theorem.

Theorem 2. *For all fixed $0 < d < 1$ and $0 < \delta < 1$, we have the following:*

$$R(D) - R^* \leqslant \max\left\{\frac{1}{1-\delta}, \frac{d}{\delta(1+d)}\right\}\left\{R_{\phi_d}(f) - R^*_{\phi_d}\right\};$$

For $\delta = 0$ and $d = 1$, we have that $R(D) - R^* \leqslant R_{\phi_d}(f) - R^*_{\phi_d}$.

Theorem 2 connects the generalization error with the surrogate loss and implies Fisher consistency, i.e., for a series of f_n, if $R_{\phi_d}(f_n) \to R^*_{\phi_d}$, then $R\left(\widehat{D}_n\right) \to R^*$, where $\widehat{D}_n = D(\boldsymbol{X}; f_n, \delta)$. Theorem 2 also provides guidance on selecting δ. For $0 < d < 1$, by solving the equation $1/(1-\delta) = d/\delta(1+d)$, the optimal choice of δ is $d/(1+2d)$, which leads to a minimal factor of $(1+2d)/(1+d)$ between $R_{\phi_d}(f) - R^*_{\phi_d}$ and $R(D) - R^*$. In this work, we specify d as a pre-specified parameter; in practice, one may specify d as a function of $\boldsymbol{X}$, which leads to a (optimal) choice of δ depending on $\boldsymbol{X}$. Through Theorem 2, we bound the generalization error $R\left(\widehat{D}\right) - R^*$ through a bound for $R_{\phi_d}(\widehat{f}_d) - R^*_{\phi_d}$.

To bound the generalization error, we decompose $R_{\phi_d}(\widehat{f}_d) - R^*_{\phi_d}$ into the estimation error and

approximation error, i.e.,

$$R_{\phi_d}(\widehat{f}_d) - R^*_{\phi_d} \leqslant R_{\phi_d}(\widehat{f}_d) + \lambda_n \left\|\widehat{f}_d\right\|^2_{\mathcal{H}} - \inf_{f\in\mathcal{H}} \left\{R_{\phi_d}(f) + \lambda_n \|f\|^2_{\mathcal{H}}\right\} + \inf_{f\in\mathcal{H}} \left\{R_{\phi_d}(f) + \lambda_n \|f\|^2_{\mathcal{H}} - R^*_{\phi_d}\right\}.$$

The first term $R_{\phi_d}(\widehat{f}_d) + \lambda_n \left\|\widehat{f}_d\right\|^2_{\mathcal{H}} - \inf_{f\in\mathcal{H}} \left\{R_{\phi_d}(f) + \lambda_n \|f\|^2_{\mathcal{H}}\right\}$ is the estimation error due to the randomness of the samples; the second term $a(\lambda_n) := \inf_{f\in\mathcal{H}} \left(\lambda_n \|f\|^2_{\mathcal{H}} + R_{\phi_d}(f) - R^*_{\phi_d}\right), \lambda_n \geqslant 0$, is the approximation error due to the constraint $f \in \mathcal{H}$. In this work, we adopt the Gaussian RBF kernels and propose generalized geometric conditions that allow us to derive the approximation error. Our generalized geometric conditions are generalizations of the geometric condition in Steinwart et al. (2007) for classification problems. Different from the geometric condition in Steinwart et al. (2007), our results focus on learning an ITR with partial identification bounds and a deferred option. Our proposed conditions do not require any kind of smoothness assumptions on the distribution or density functions. Instead, in the proposed generalized geometric conditions, we define a generalized geometric noise exponent $\alpha > 0$, which is related to on the concentrations of the covariates near the decision boundaries. In classification problems, the approximation error at each covariate value $\boldsymbol{x}$ contributes differently to the approximation error $a(\lambda_n)$; and the covariate value $\boldsymbol{x}$ closer to the decision boundaries contribute more to the approximation error. Thus, by the proposed generalized geometric conditions, we can control the concentration near these boundaries and thus establish the approximation errors. We also show that the Hölder continuity and bounded density functions are sufficient conditions for the proposed generalized geometric conditions. Thus, the generalized geometric conditions are weaker than the smoothness assumptions on the distribution or density functions. The details can be found in the online Supplementary Materials.

Theorem 3. *Let $\sigma_n > 0$ be the bandwidth of Gaussian RBF kernels, $\mathcal{X}$ be the closed unit ball in $\mathbb{R}^p$, P_X is a distribution on $\mathcal{X}$. Furthermore, P_X has generalized geometric noise exponent $\alpha > 0$.*

Then there is a constant $C_d > 0$ depending only on the dimension p such that for all $\lambda_n > 0$, we have $a(\lambda_n) \leqslant C_d(\sigma_n^p \lambda_n + (2d)^{\alpha p/2} \sigma_n^{-\alpha p})$.

Theorem 3 provides an upper bound for the approximation error using Gaussian RBF kernels. From Theorem 3, the optimal choice of σ_n to minimize the upper bound for the approximation error is $\lambda_n^{-1/(1+\alpha)p}$. Under this choice, the approximation error is bounded by $a(\lambda_n) \leqslant C_d(1 + (2d)^{\alpha p/2})\lambda_n^{\alpha/(1+\alpha)} \leqslant 2C_d\lambda_n^{\alpha/(1+\alpha)}$, which is independent of the dimension p. In our simulation and real data example, we use the cross-validation to choose σ_n.

Finally, we provide an upper bound for the generalization error $R\left(\widehat{D}\right) - R^*$. Unlike the traditional ERM in Zhao et al. (2012, 2019); Pu and Zhang (2021b), deriving the estimation error for the augmented ERM is challenging due to non-smooth transformation in constructing $B_+(\boldsymbol{X}, Y, A, Z)$ and $B_-(\boldsymbol{X}, Y, A, Z)$. To account for the non-smooth transformations, we introduce new margin conditions for partial identification.

Definition 1. *We say that P_X has margin condition exponent $\gamma > 0$ if there exists a constant $C_\gamma > 0$ such that for $t \leqslant t_0 \in (0, 1)$,*

$$\begin{aligned} P\left\{0 \neq |\min\left\{Q_1(\boldsymbol{X}), \cdots, Q_K(\boldsymbol{X})\right\}| \leqslant t\right\} &\leqslant C_\gamma t^\gamma, \quad \sup_{k,k'} P\left\{0 \neq |Q_k(\boldsymbol{X}) - Q_{k'}(\boldsymbol{X})| \leqslant t\right\} \leqslant C_\gamma t^\gamma \\ P\left\{0 \neq |\max\left\{P_1(\boldsymbol{X}), \cdots, P_K(\boldsymbol{X})\right\}| \leqslant t\right\} &\leqslant C_\gamma t^\gamma, \quad \sup_{k,k'} P\left\{0 \neq |P_k(\boldsymbol{X}) - P_{k'}(\boldsymbol{X})| \leqslant t\right\} \leqslant C_\gamma t^\gamma. \end{aligned}$$

These conditions characterize the concentration near the boundaries $\{\min\{Q_1(\boldsymbol{X}), \cdots, Q_K(\boldsymbol{X})\} = 0\}$, $\{\min\{P_1(\boldsymbol{X}), \cdots, P_K(\boldsymbol{X})\} = 0\}$, $\{Q_k(\boldsymbol{X}) = Q_{k'}(\boldsymbol{X})\}$, and $\{P_k(\boldsymbol{X}) = P_{k'}(\boldsymbol{X})\}$. The constant C_γ can be different for different boundaries; however, for the simplicity of the notations, we use a common C_γ in the definition. Notice that we allow positive mass on these boundaries, and thus these conditions are not restrictive. Further, when $P_k(\boldsymbol{X})$'s, $Q_k(\boldsymbol{X})$'s, and their differences have bounded p.d.f. with respect to the Lebesgue measure, then P_X has a margin condition exponent at least 1, i.e., $\gamma \geqslant 1$; when P_X are bounded away from these boundaries, P_X has margin condition exponent $\gamma = +\infty$. It is

also possible to have margin condition exponent smaller or larger than 1. For example, when $P_k(\boldsymbol{X})$'s, $Q_k(\boldsymbol{X})$'s, and their differences have density functions proportional to $|t|^2$ near 0 (for $t \leqslant t_0$), then P_X has a margin condition exponent $\gamma = 2$; and when $P_k(\boldsymbol{X})$'s, $Q_k(\boldsymbol{X})$'s, and their differences have density functions proportional to $|t|^{-1/2}$ near 0, then P_X has a margin condition exponent $\gamma = 1/2$.

We further assume the following assumptions.

C1) There exists $0 < \epsilon < 1$ such that $\epsilon \leqslant \pi_z(\boldsymbol{x}) \leqslant 1 - \epsilon$ for all $\boldsymbol{x} \in \mathcal{X}$.

C2) Suppose that for some $\beta_\pi, \beta_p > 0$,

$$\sup_{\boldsymbol{x},z} |\widehat{\pi}_z(\boldsymbol{x}) - \pi_z(\boldsymbol{x})| = O_p(n^{-\beta_\pi}), \quad \sup_{\boldsymbol{x},z,a,y} \left|\widehat{p}_{y,a|z}(\boldsymbol{x}) - p_{y,a|z}(\boldsymbol{x})\right| = O_p(n^{-\beta_p}).$$

Assumption C1) is the positivity assumptions that are commonly adopted in causal inference literature. Assumption C2) specifies the uniform convergence rates of the nuisance estimations. These assumptions can be satisfied if $\pi_z(\boldsymbol{x})$ and $p_{y,a|z}(\boldsymbol{x})$ are parametric or non-parametric under certain smoothness conditions. For example, if $p_{y,a|z}(\boldsymbol{x})$ and $\pi_z(\boldsymbol{x})$ are estimated by regression spline estimators and are known to be p_d-dimensional (low-dimensional) by design, then we have $\beta_s = \beta_p = s/(2s + p_d)$ if $\pi_z(\boldsymbol{x})$ and $p_{y,a|z}(\boldsymbol{x})$ belong to the Hölder class with a smoothness parameter larger than s (Newey 1997). Under these assumptions, we have the following theorem.

Theorem 4. *Under Assumptions C1) and C2), let P_X has generalized geometric noise exponent α and margin condition exponent γ, then for any $h > 0$, $0 < v < 2$, with $\sigma_n = \lambda_n^{-1/(1+\alpha)p}$, we have*

$$\begin{aligned} & R\left(\widehat{D}\right) - R^* \\ = \; & O_p\left(\lambda_n^{-\frac{v}{v+2} - \frac{(2-v)(1+h)}{(2+v)(1+\alpha)}} n^{-2/(2+v)} + \lambda_n^{-\frac{1}{2} - \frac{(2-v)(1+h)}{4(1+\alpha)}} n^{-1/2} + n^{-(1+\gamma)\beta_p} \lambda_n^{-1/2} + n^{-(\beta_\pi + \beta_p)} \lambda_n^{-1/2} + \lambda_n^{\frac{\alpha}{1+\alpha}}\right). \end{aligned}$$

Compared with the results in Pu and Zhang (2020), our theoretical result takes the approximation error into account. In addition, Theorem 4 shows that the risk of the estimator converges to R^* with a

faster convergence rate than traditional ERM, especially results in Pu and Zhang (2020). Considering the terms involving the convergence rates of nuisance parameters, i.e., β_p and β_π, our result shows that convergence rate of augmented ERM is bounded by $n^{-(1+\gamma)\beta_p}\lambda_n^{-1/2} + n^{-(\beta_\pi+\beta_p)}\lambda_n^{-1/2}$; in Pu and Zhang (2020), the terms involving the convergence rates of nuisance parameters is $\left(n^{-\beta_\pi} + n^{-\beta_p}\right)\lambda_n^{-1/2}$, which is slower than ours.

5 Simulations

We conduct simulations to examine the performance of the EIV-ITR and the proposed augmented ERM to estimate EIV-ITR.

5.1 Settings

We generate the data following the procedure below. First, we generate the covariate $\boldsymbol{X}$, where each coordinate follows an independent uniform distribution on the interval $[-1, 1]$. We then generate an unobserved confounder, U, which follows a uniform distribution on $[-1, 1]$. Next, we generate an IV, Z, which follows a binomial distribution with a success probability of 0.5. This IV is independent of both the covariate $\boldsymbol{X}$ and U. To generate the observed treatment assignment, we utilize a logistic model that involves $\boldsymbol{X}$, U, and Z. Specifically, we have:

$$\log\left\{\frac{P(A=1 \mid \boldsymbol{X}, U, Z)}{P(A=-1 \mid \boldsymbol{X}, U, Z)}\right\} = 8Z + X_1 - 7X_2 + \xi(1+X_1)U,$$

where ξ is the parameter that controls the strength of confounding in the simulation settings. Then, we generate potential outcomes $Y(a)$'s following logistic models involving $\boldsymbol{X}$, U, and Z:

$$\log\left\{\frac{P(Y_a=1 \mid \boldsymbol{X}, U, Z)}{P(Y_a=-1 \mid \boldsymbol{X}, U, Z)}\right\} = g_1(\boldsymbol{X}, U) + g_2(\boldsymbol{X}, U, a).$$

We have two specifications for $g_1(\boldsymbol{X}, U)$ and $g_2(\boldsymbol{X}, U, a)$, which lead to a total of four simulation settings:

Setting $(1,1)$: $g_1(\boldsymbol{X}, U) = 1 - X_1 - X_2 + \xi U$; $g_2(\boldsymbol{X}, U, a) = 0.442a(1 - X_1 - X_2 + \xi U)$.

Setting $(1,2)$: $g_1(\boldsymbol{X}, U) = 1 - X_1 - X_2 + \xi U$; $g_2(\boldsymbol{X}, U, a) = a(4 + X_1 - 0.25\xi X_1^2 - 2 \times 1\{X_2 \leqslant 0\})$.

Setting $(2,1)$: $g_1(\boldsymbol{X}, U) = 1 - X_1^2 + X_2^2 + \xi X_1 X_2 U$; $g_2(\boldsymbol{X}, U, a) = 0.442a(1 - X_1 - X_2 + \xi U)$.

Setting $(2,2)$: $g_1(\boldsymbol{X}, U) = 1 - X_1^2 + X_2^2 + \xi X_1 X_2 U$; $g_2(\boldsymbol{X}, U, a) = a(4 + X_1 - 0.25\xi X_1^2 - 2 \times 1\{X_2 \leqslant 0\})$.

In comparison to Settings $(1,1)$ and $(2,1)$, Settings $(1,2)$ and $(2,2)$ have a higher percentage of samples where the optimal treatment decision is identifiable. For each simulation setting, we set the sample size in the training dataset n to 300 and 500, and the confounding parameter ξ is set to 0.5 (representing weak confounding) and 1.5 (representing strong confounding).

To assess the performance, we generate a testing dataset consisting of 1000 samples. We compute three metrics on the testing dataset for comparison purposes: 1) the (worst) total loss in value under the estimated ITR, i.e., $E\left[\sup_{\Delta(\boldsymbol{X}) \in [L(\boldsymbol{X}), U(\boldsymbol{X})]} \widetilde{TLV}(\widehat{D}, \boldsymbol{X}; \Delta, d)\right]$; 2) the average outcome under the estimated ITR on the non-deferred subgroup, i.e., $E[Y_{\widehat{D}} \mid \widehat{D}(\boldsymbol{X}) \neq 0]$; and 3) the proportion of people receiving the deferred option, i.e., $P(\widehat{D}(\boldsymbol{X}) = 0)$. For each setting, the entire procedure including generating a training dataset, fitting different methods, and evaluating on a testing dataset, is repeated 200 times. In the online Supplementary Materials, we have additional simulations when outcomes are continuous.

5.2 Results

We compare the augmented ERM (A-ERM) to three alternative methods. The first method utilizes traditional ERM to estimate D_d^*. By comparing our approach to ERM, we can highlight the advantages of using A-ERM for estimating D_d^*. The second method, referred to as 'Wald', emulates a common usage of a deferred option. This method involves two steps: 1) estimating the decision

under model assumptions, where CATEs are point identifiable; 2) defer predictions with CATEs close to 0. Specifically, the Wald method estimates $\Delta_Y := E[Y \mid Z = 1, \boldsymbol{X}] - E[Y \mid Z = -1, \boldsymbol{X}]$ and $\Delta_A := E[A \mid Z = 1, \boldsymbol{X}] - E[A \mid Z = -1, \boldsymbol{X}]$ using kernel regressions and treats $\widehat{\Delta}_Y/\widehat{\Delta}_A$ as the estimates of the CATEs. Under assumptions in Cui and Tchetgen Tchetgen (2021); Qiu et al. (2021), this method yields a consistent estimator; however, in our simulations, Assumption 7 in Cui and Tchetgen Tchetgen (2021) and Assumption 5(b) in Qiu et al. (2021) are violated, and thus this method may lead to a biased estimation. To accommodate these possible violations of the assumptions, a cutoff value is then chosen to construct a deferred option for the Wald method. Individuals whose absolute CATE estimates fall below the cutoff are deemed to have their treatment decisions deferred. The underlying rationale is that small estimates of the CATEs may be susceptible to violation of the assumptions. To ensure a fair comparison, we select the cutoff such that the proportion of deferred patients matches that of the proposed method and compare the value on the non-deferred patients. Comparing our approach to the Wald method allows us to highlight the benefits of connecting the deferred option with identification uncertainty. The third method is the method in Pu and Zhang (2020) where no deferred option is employed; when $d = 1$, the proposed framework and the method proposed in Pu and Zhang (2020) yield the same IV-optimal ITR (without deferred options). Thus, it would be interesting to investigate whether the proposed augmented empirical risk minimization (A-ERM) method can still outperform the approach proposed in Pu and Zhang (2020) in this case. Furthermore, for each setting, we vary the choice of d from 0.2 to 1 in increments of 0.2. This allows us to evaluate the performance of the methods under different proportions of individuals choosing the deferred option.

Figure 4 provides a summary of the comparison metrics for each method in Settings $(1, 2)$ and $(2, 2)$ across 200 repeats. The online Supplementary Materials include simulation results in Settings $(1, 1)$ and $(2, 1)$, and additional metrics to evaluate the performance of deferral. Based on Figure 4, it can be observed that the proposed A-ERM yields ITRs with the smallest loss in value and the highest

average outcomes on the non-deferred subgroup. Regarding the proportions of individuals receiving the deferred option, ERM tends to employ more deferred options compared to the proposed method, although the average outcomes on the non-deferred subgroup are lower. While the Wald method defers the same number of patients as A-ERM, the average outcomes on the non-deferred subgroup are lower than A-ERM. This indicates that the Wald method may mistakenly defer some patients with high true CATE values, implying that the Wald use of the deferred option might not effectively mitigate decision errors when the model assumption is likely to be violated. Our proposed framework, EIV-ITR, is designed to defer treatment decisions for individuals whose recommendations are prone to identification uncertainty. However, the Wald method may not necessarily defer these individuals. Consequently, our proposed framework offers a more robust approach for incorporating the deferred option to deal with possible identification uncertainty. Figure 5 compares with the proposed method in Pu and Zhang (2020), and shows that the A-ERM approach yields the lowest loss in value and the highest average outcomes under the estimated ITRs. This result demonstrates the effectiveness of A-ERM in mitigating finite sample bias associated with non-parametric or machine learning methods, and achieving a faster convergence rate in terms of generalization error.

6 Application

We apply the EIV-ITR framework to the PCPT. The goal is to evaluate who is at an elevated risk of developing high-grade prostate cancer when using finasteride, and to develop individualized treatment recommendations accounting for the actual medication that patients take. The outcome of interest is whether the patient will develop high-grade prostate cancer ($Y = -1$) vs. developing low-grade prostate cancer or staying healthy ($Y = 1$). To make decisions based on the actual effect of finasteride rather than the intent-to-treat effect, we use the random assignment as an IV, i.e., if a patient is assigned to finasteride arm, we have that $Z = 1$; otherwise, $Z = -1$. In the finasteride arm, whether a patient takes finasteride is determined by adherence to the study protocol. Following Pinsky et al.

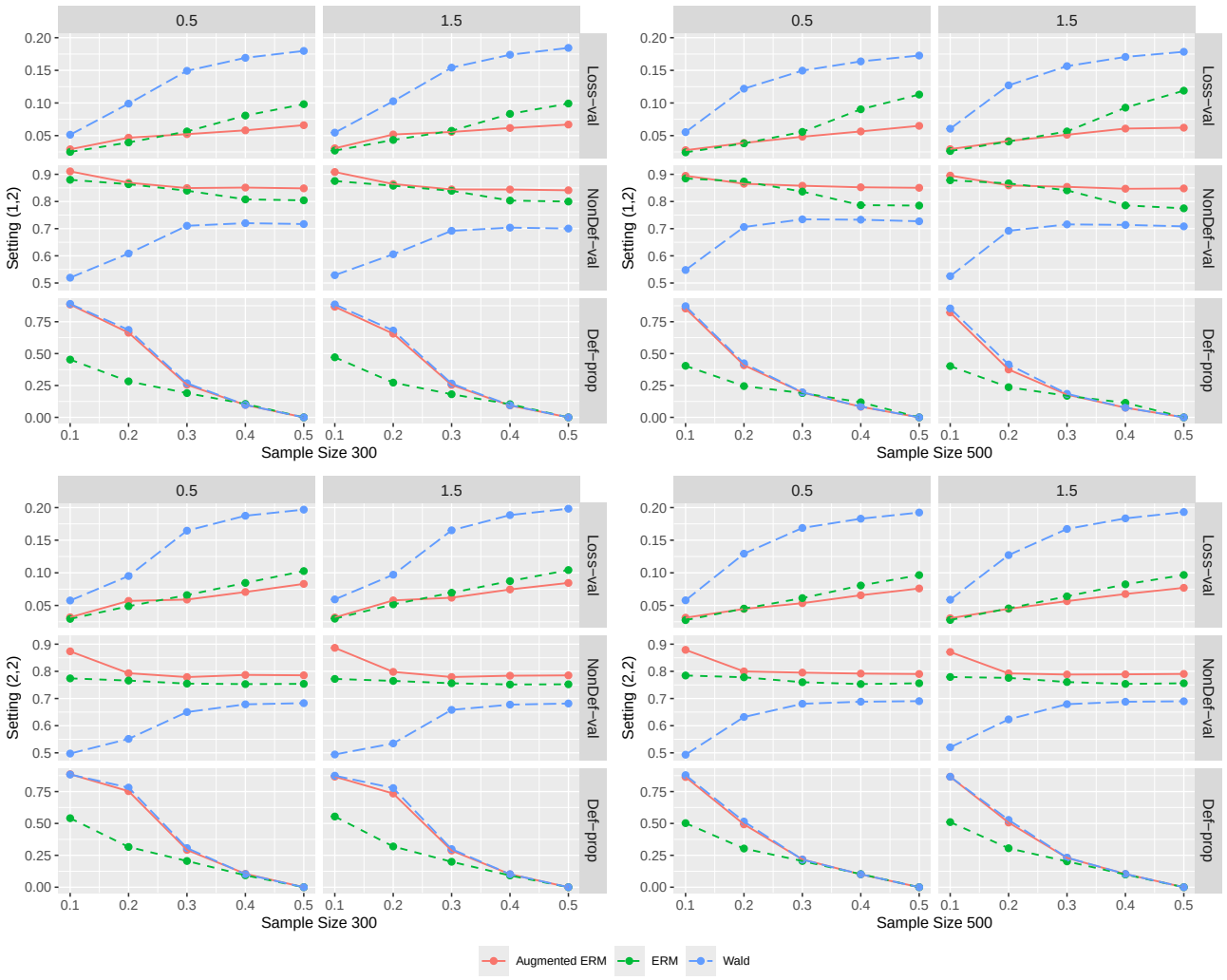


Figure 4: Simulation results for Settings $(1, 2)$ and $(2, 2)$ over 200 repeats.

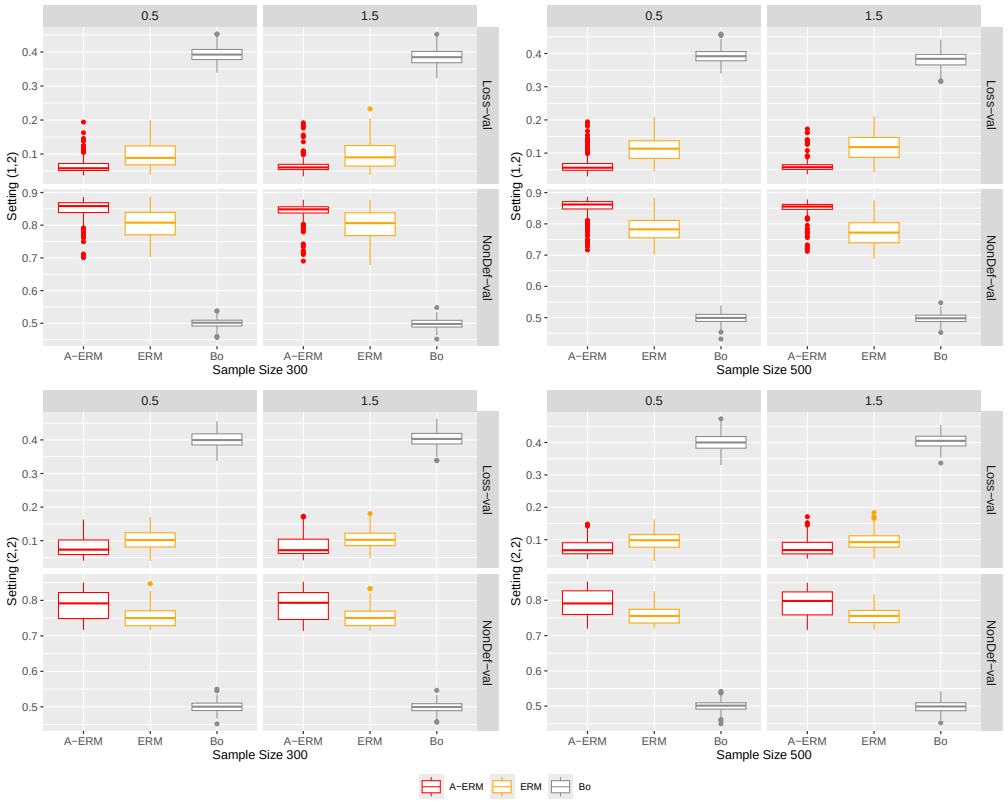


Figure 5: Compare A-ERM, ERM, and the method in Pu and Zhang (2020). The A-ERM approach achieves the best performance among all methods.

(2021), adherence is assessed by pill counts at 6 and 12 months of the study. If both pill counts are greater than 80%, we consider the patient adherent; otherwise (if any pill counts is below 80% or if there are missing pill counts), we consider the patient non-adherent. If a patient is assigned to finasteride arm and is adherent, then $A = 1$; otherwise, we consider $A = -1$.

We split the entire data into a training dataset and a testing dataset with a sample size ratio of $1 : 9$. Then, we estimate ITRs with a deferred option using the training dataset and use the estimated ITRs to make decisions for patients in the testing dataset. To examine the estimated ITRs under different risks of taking the deferred option, we vary d from 1 to 0.11. For each choice of d, if the learned ITR $\widehat{D} = 1$, then finasteride is suggested by the learned ITR; if the learned ITR $\widehat{D} = -1$, then finasteride is NOT suggested by the learned ITR; if the learned ITR $\widehat{D} = 0$, then making any treatment decisions leads to a higher risk than following the deferred option due to identification uncertainty.

Figure 6 illustrates the treatment decisions for the patients in the testing dataset using the estimated ITRs with varying discount factors d. To plot covariate information in Figure 6, we use the entire data and conduct the principal component analysis; each point in each subplot represents the projection of the covariates from one patient onto the first two principal components. When $d = 0.11$, i.e., the risk of following a deferred option is relatively small, most patients will receive the deferred option. When $d \geqslant 0.14$, there seems to be a stable subgroup (showing in blue) suggesting an elevated risk if taking finasteride. These results indicate that although most patients are affected by the non-adherence issue and their optimal decisions are susceptible to the identification uncertainty, there seems to be a subgroup for whom it is indeed too risky to take finasteride. For others who are suggested to take finasteride or the deferred option by the estimated ITRs, they may follow other clinical considerations to make treatment decisions, e.g., take finasteride if showing a reduced risk of developing all grades of prostate cancer (see an example in the online Supplementary Materials).

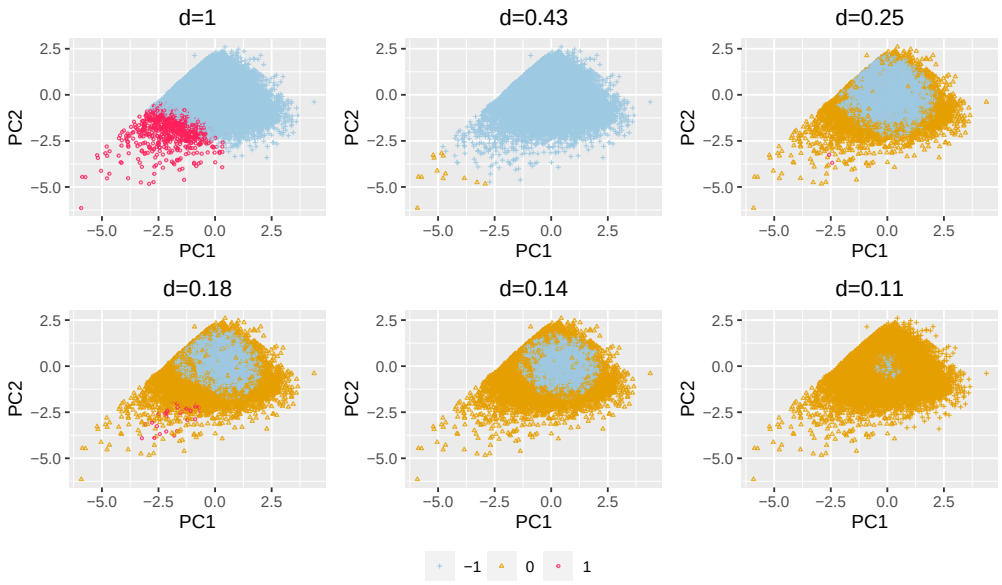


Figure 6: Decisions of the estimated ITRs with varying discount factor d.

7 Discussion

In this work, we propose a novel framework that simultaneously estimates the optimal ITR for the subgroup with identifiable optimal treatment decisions and incorporates a deferred option for the subgroup where optimal treatment decisions may not be identified. The EIV-ITR framework allows users to control the trade-off between the risk of making incorrect decisions due to identification uncertainty and the risk of following alternative strategies through a deferred option. Our results demonstrate that it is sub-optimal to make decisions with a deferred option based on point identification under a set of possibly invalid assumptions. Instead, our framework incorporates the deferred option to account for possible impact of identification uncertainty under more relaxed assumptions. We also propose a novel A-ERM to estimating the EIV-ITR.

There are multiple possible extensions of this work. Our framework can be extended to general cases with partial identification of CATE, e.g., partial identification due to measurement errors/sensitivity analysis. For example, we can accommodate non-binary or continuous IVs as long as the partial identification bound can be established and consistently estimated. In this extension, in

terms of the estimation procedure, we further require the partial identification bound following the format as $L(\boldsymbol{x}) := \max_{k\{1,\cdots,K\}}\{P_k(\boldsymbol{x})\} \leqslant \Delta(\boldsymbol{x}) \leqslant \min_{k\{1,\cdots,K\}}\{Q_k(\boldsymbol{x})\} =: U(\boldsymbol{x})$, and $P_k(\boldsymbol{x})$'s and $Q_k(\boldsymbol{x})$'s are differentiable/sub-differentiable based on the conditional means of $(Y, A) \mid (Z, \boldsymbol{X})$; these ensure the validity of the A-ERM algorithm. One of the examples is the inequality in Richardson and Robins (2014) for binary outcome and treatment under a discrete IV (M values/states in IV), which provides an inequality following the format above with $K = 2M^2$. However, the partial identification bound may not always follow the format as required above, especially with continuous IVs, and thus the proposed A-ERM may not be directly applied. Th key challenge is that these partial identification bounds often involves integral over continuous values of treatment/instrument (Heckman and Vytlacil 1999; Mogstad et al. 2018; Tan et al. 2026); and thus how to remove the bias due to nuisance estimation is not straightforward and requires more detailed and involving analysis. Our framework and estimation approach in this work lay a solid foundation for these developments.

Extending the methodology to continuous treatments is also an important future direction but requires substantially new methodological development. In our application, adherence measured by pill counts or self-report is a noisy proxy for true drug exposure, and our current framework assumes that the observed treatment is measured without error. One natural question is whether the adherence should be treated as a variable with misclassifications or a continuous variable. Extending the proposed framework to accommodate treatment misclassifications would require detailed characterization of the misclassifications mechanism and related partial identification results. In addition, although treating adherence as a continuous treatment is an interesting direction, in our setting this is more challenging than a standard continuous-treatment problem because treatment is instrumented by a binary randomized assignment. This extension would require deriving new partial identification results for entire dose-response curve with or without strong shape constraint (Vohra and Goldin 2024; Ishihara 2021); another challenge is how to incorporate the loss of unidentified optimal dose decision with the existing methods in optimal dose finding through classification frameworks since it

fundamentally changes the optimal treatment decision to the problem of optimal dose finding. We will leave these for future investigation.

Supplemental Materials

Proofs of all theorems and additional simulation results are contained in the online Supplemental Materials. Data Sets in Application cannot be shared publicly due to data access agreements. Computer/Source Code including simulation codes will be shared on public repository, e.g., GitHub.

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